Peoples Empowerment Group
ISBM COLLEGE OF ENGINEERING, NANDE, PUNE
Department of Artificial Intelligence and Machine Learning
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PROJECT SYNOPSIS

Project Title:

PROPCHAIN-AI: BLOCKCHAIN-BASED AI SYSTEM FOR PROPERTY ANALYSIS

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Problem Definition

Property transactions in India require examination of multiple documents such as registered sale deeds, previous title documents, 7/12 extracts, 8A records, Ferfar/mutation entries, Property Cards, Index records, encumbrance information, tax receipts, RERA records, succession documents and other supporting records. These documents may come from different sources, use different formats or languages, and may contain inconsistent names, dates, areas, survey/Gat/CTS numbers or ownership references. Manual comparison is time-consuming and can cause important discrepancies to be overlooked. The proposed BHOOMI CHAIN AI system provides an AI-assisted due-diligence platform that uses OCR, NLP, document classification, cross-document comparison and ownership-graph analysis to identify potential inconsistencies and missing links. Blockchain is used to store cryptographic document fingerprints and verification events, providing a tamper-evident audit trail. The system assists users in property due diligence but does not establish legal title or replace government records and professional legal advice.

Key Objectives

- Classify uploaded Indian property documents such as Sale Deeds, Gift Deeds, Mortgage Deeds, Lease Deeds, 7/12, 8A, Ferfar and Property Cards.
- Extract names, dates, property identifiers, area, transaction details and other relevant entities using OCR and NLP.
- Normalize and compare property information across multiple documents to detect potential discrepancies.
- Construct a chronological ownership/transaction chain and identify missing or conflicting links.
- Apply a configurable Central-law and Maharashtra-specific rule engine to generate explainable due-diligence findings.
- Generate a system-defined risk indicator with evidence, confidence and recommended verification steps.
- Store document hashes and verification events on blockchain without placing sensitive document contents on-chain.
- Provide a clear, auditable property passport and downloadable due-diligence report.

Project Domain

PROPCHAIN-AI lies at the intersection of Artificial Intelligence and Machine Learning, Natural Language Processing, Document Intelligence, Blockchain Technology, E-Governance, Property/Land Information Systems, Backend Web Development, Database Management and Cybersecurity. The system combines OCR and NLP for document understanding, blockchain for document integrity, and a jurisdiction-aware rules engine for property due diligence, with Maharashtra as the initial implementation state.

Literature Survey

Ref. No.	Title of the Paper	Author / Publication / Year	Technique Used	Summary
1	Land records on Blockchain for implementation of Land Titling in India	Vinay Thakur et al. / International Journal of Information Management / 2020	Blockchain, smart contracts, land records	Discusses fragmented Indian land records, transparency and security issues, and proposes blockchain-based land-record and land-titling architecture.
2	Blockchain as a tool to facilitate property rights protection in the Global South: lessons from India's Andhra Pradesh state	Third World Quarterly / 2022	Blockchain, land administration, property rights	Studies an Indian blockchain land-registry initiative and discusses benefits, adoption barriers and institutional requirements.
3	A systematic review of the institutional success factors for blockchain-based land administration	Land Use Policy / 2022	Systematic review, blockchain land administration	Shows that successful blockchain land administration depends not only on technology but also on regulative, normative and institutional support.
4	Design and Implementation of a Geospatially Enabled Blockchain Application for Land Record Management	ISPRS Annals / 2025	Blockchain, GIS, hashing, smart contracts	Combines geospatial data with blockchain for land-record integrity, transaction traceability and stakeholder confidence.
5	Blockchain-enabled secure land record management system with a new lightweight cryptosystem	Blockchain: Research and Applications / 2026	Blockchain, IPFS, encryption, land records	Presents a privacy-oriented blockchain land-record framework and reinforces the value of protecting sensitive land data while maintaining integrity.

Research Gap

Existing research demonstrates the usefulness of blockchain for land-record integrity and transparency, but many approaches focus primarily on ledger storage or transaction management. PROPCHAIN AI addresses an additional layer: AI-assisted understanding of heterogeneous property papers, cross-document consistency checking, ownership-chain construction and explainable risk indicators. The project also separates blockchain integrity from legal-title determination and introduces a jurisdiction-aware rule layer, beginning with Maharashtra and designed for later extension to other Indian states.

Proposed System and Methodology

The proposed system follows a hybrid architecture in which deterministic processing handles document integrity and rule evaluation, while AI/ML is used for document understanding and similarity analysis. The workflow begins with secure document upload, followed by OCR and document classification. Extracted entities are normalized into a canonical property model. The system then compares records, constructs an ownership/transaction graph, evaluates configurable legal-rule conditions and creates explainable findings. Finally, a SHA-256 fingerprint of each source file is anchored to blockchain, while the actual file remains in protected storage.

Major Modules

- User and Property Management - authentication, role-based access, property profiles and unique property IDs.
- Document Management - upload, classification, versioning, metadata extraction and secure storage.
- OCR and NLP Engine - multilingual OCR support, text cleaning, named-entity extraction and structured field mapping.
- Property Consistency Engine - comparison of names, dates, survey/Gat/CTS numbers, area, boundaries and registration information.
- Ownership Graph - chronological representation of sale, gift, mortgage, lease, partition, inheritance and other transfers.
- Legal Rule Engine - Central legal framework plus Maharashtra-specific rules and source/version tracking.
- Risk and Explainability Engine - severity, confidence, evidence, source and recommended verification for each finding.
- Blockchain Integrity Module - document hashing, verification and immutable audit events.
- Report and Property Passport - consolidated due-diligence summary, ownership timeline, findings and blockchain verification status.

Indicative Indian Legal / Regulatory Coverage

The rule engine is designed to reference current official legal and government sources rather than hard-code unsupported legal conclusions. The project scope includes the Transfer of Property Act, 1882; Registration Act, 1908; applicable stamp legislation; Indian Succession Act, 1925 and relevant succession frameworks; Hindu Succession Act, 1956 where applicable; Specific Relief Act, 1963; Limitation Act, 1963; Easements Act, 1882; Prohibition of Benami Property Transactions Act, 1988; SARFAESI Act, 2002; RERA, 2016; Land Acquisition, Rehabilitation and Resettlement Act, 2013; FEMA-related property restrictions; Bharatiya Sakshya Adhiniyam, 2023; Digital Personal Data Protection Act, 2023 and applicable rules; and relevant state/local land, tenancy, agricultural, ceiling, planning, registration and municipal rules. The initial state pack focuses on Maharashtra, including land-record and planning concepts such as 7/12, 8A, Ferfar, Property Card, Survey/Gat/CTS, ULPIN, Maharashtra Land Revenue Code, Maharashtra Stamp Act, applicable agricultural-land/ceiling restrictions, town planning and MahaRERA.

Core AI / Blockchain Workflow

1. Upload

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2. File Validation

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3. OCR

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4. Language Detection

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5. Document Classification

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6. Entity Extraction

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7. Data Normalization

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8. Cross-Document Comparison

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9. Ownership Graph Construction

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10. Legal Rule Evaluation

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11. Risk Findings Generation

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12. Human Review

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13. SHA-256 Hashing

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14. Blockchain Verification

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15. Property Passport / Final Report

Responsible AI and Legal Safeguards

- The system reports potential findings and documentary inconsistencies; it does not declare legal ownership or replace a registrar, government record or advocate.
- AI confidence is distinct from legal certainty. Unknown, not checked, not applicable and source unavailable states are supported.
- Sensitive documents are stored off-chain; only cryptographic fingerprints and minimum audit metadata are placed on blockchain.
- Every significant finding records evidence, source, document/page reference where available, rule version and verification status.

Scope of the Project

- Initial implementation for Maharashtra, with an extensible architecture for other Indian states and Union Territories.
- Analysis of commonly encountered property papers including Sale Deeds, Gift Deeds, Mortgage/Lease documents, Partition/Release documents, Agreements to Sale, Wills/POA, 7/12, 8A, Ferfar, Property Card, encumbrance records and selected supporting documents.
- AI-assisted detection of owner/name inconsistencies, property-identifier mismatches, area discrepancies, chronology conflicts, missing predecessor documents, potential encumbrance conflicts and ownership-chain gaps.
- Optional screening modules for RERA/project information, court/litigation documents, land-use/planning indicators, inheritance/succession indicators and tax/compliance clues, subject to source availability.
- Blockchain-based file-integrity verification and auditable analysis history.
- Generation of a Property Passport and due-diligence report for academic demonstration using synthetic/anonymized data.

Software Requirements

- Programming Language: Python
- Frontend: Next.js / React / TypeScript / Tailwind CSS
- Backend: FastAPI with RESTful APIs
- Database: PostgreSQL
- AI/ML: OCR engine, spaCy/Transformers, sentence-transformers, scikit-learn and PDF processing libraries
- Blockchain: Solidity, Web3.py and a local/private Ethereum-compatible test network or permissioned ledger for the prototype
- Storage/Security: encrypted object storage, JWT-based authentication, role-based access control and audit logs
- Development Tools: Visual Studio Code, Git, API testing tools and browser-based testing environment
- Operating System: Windows / Linux / macOS

Hardware Requirements

- Processor: Intel Core i5 / AMD Ryzen 5 or equivalent and above recommended
- RAM: Minimum 8 GB; 16 GB recommended for OCR, NLP and blockchain development
- Storage: Minimum 20 GB free space for project tools, datasets, models and blockchain environment
- GPU: Optional; not mandatory for baseline implementation
- Internet connection: Required for package installation, optional cloud AI services and access to permitted public information sources

Expected Outcomes

- A working web application that converts uploaded property papers into structured property information.
- A visual ownership/transaction timeline and graph that highlights missing or conflicting links.
- Explainable risk indicators supported by document evidence and configurable legal-rule references.
- Blockchain-based verification showing whether a digital document matches its previously registered fingerprint.
- A downloadable Property Due-Diligence Report and Property Passport suitable for academic demonstration.

Limitations

The prototype cannot guarantee legal title, detect undisclosed facts outside accessible records, replace professional conveyancing or independently establish the absence of litigation. Government sources may have access controls, delays or jurisdiction-specific coverage. Legal rules can change; therefore, the system must maintain rule versions and verification dates.

Project Guide	Project Co-ordinator	Head of Department
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